Case Report

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Case report: a rare case of cutaneous lymphangitic carcinomatosis from colonic signet ring cell carcinoma

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Introduction and importance: Cutaneous metastasis from internal malignancies is rare, and cutaneous lymphangitic carcinomatosis (CLC) represents an even rarer subset characterized by tumor cell infiltration into the dermal and subcutaneous lymphatics. In this case report, we describe an unusual presentation of CLC originating from colonic signet ring cell carcinoma and highlight the importance of maintaining a high index of suspicion for timely diagnosis.

Case presentation: A 24-year-old male patient with no prior history of malignancy presented to us with a 10-month history of progressive, painless bilateral lower limb and scrotal swelling accompanied by significant weight loss. Skin biopsy with immunohistochemical analysis confirmed cutaneous lymphangitic carcinomatosis secondary to metastatic colonic signet ring cell carcinoma.

Clinical discussion: The incidence of cutaneous metastases is around 5.3–10.4% and only 4% of all cutaneous metastatic disease originates from colorectal cancers. Cutaneous lymphangitis carcinomatosis is an exceedingly rare presentation occurring in less than 5% of cases. The prognosis for patients with cutaneous lymphangitis carcinomatosis and cutaneous metastases in general is poor, with demise typically occurring within months after diagnosis of skin involvement. A high index of suspicion is required to correctly diagnose cutaneous lymphangitis carcinomatosis. Skin biopsy and immunohistochemical staining are essential for the diagnosis of cutaneous metastasis and determining the primary cancer.

Conclusion: Given its infrequency and diagnostic challenges, this case underscores the importance of recognizing suspicious skin lesions as possible harbingers of undiagnosed visceral malignancy in patients with or without a history of cancer.

Keywords: case report, colorectal carcinoma, cutaneous lymphangitic carcinomatosis, cutaneous metastasis, signet-ring cell carcinoma

Introduction

Cutaneous metastasis of internal malignancies is rare, occurring in 0.7–9.0% of all malignancies[1]. Those arising from adenocarcinomas of the colon and rectum are even more rare and typically signify widespread disease with a poor prognosis[1]. It most often occurs late in the course of disease but also may be the presenting sign of underlying cancer[2].

Signet-ring cell carcinoma (SRCC) is a mucin-secreting adenocarcinoma that contains abundant intracytoplasmic mucin that pushes the nucleus to the periphery[3]. Signet-ring cutaneous metastasis is extremely rare and indicates a poor prognosis[3].

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HIGHLIGHTS

- Cutaneous lymphangitic carcinomatosis (CLC) is a rare form of cutaneous metastatic disease.
- Signet ring cell cutaneous metastases arising from colorectal carcinomas are even rarer and signify widespread disease with a poor prognosis.
- No single clinical presentation reliably diagnoses any form of cutaneous metastasis; skin biopsy and immunohistochemical staining are essential for diagnosis.

Cutaneous lymphangitic carcinomatosis (CLC) is a rare form of cutaneous metastatic disease that is caused by dermal and subcutaneous lymphatic invasion by tumor cells[4]. It is a rare form of cutaneous metastasis with a variety of clinical presentations that can alert healthcare providers to cancer recurrence or even previously undiagnosed malignancy[5].

This case report has been reported in line with the SCARE checklist[6].

Case presentation

A 24-year-old male patient presented to us with progressive and painless swelling of bilateral lower limbs and scrotum of 10 months duration. The swelling initially started from the right groin area and later progressed to involve the lower limbs, scrotum, abdomen, and back areas. He also has associated loss of appetite and significant but unquantified weight loss. He has no history of similar illness in the family. He has no past medical or surgical illness and he is not on any medication.

The patient was initially managed at another hospital with courses of oral antibiotics for empirical diagnoses of lymphogranuloma venereum and lymphatic filariasis. Despite this, his clinical condition continued to worsen, leading to referral to our center for further evaluation and management.

Upon presentation to our center, on physical examination, there was pitting edema of the bilateral lower limbs with thickened skin and *peau d'orange* appearance. There were grossly swollen scrotum and penis with cobbling of the overlying skin (Fig. 1). There was also non-pitting edema of the lower chest and abdomen with erythematous overlying skin. And, there is dullness to percussion and decreased air entry bilaterally on the lower one-third lung fields.

Given the clinical picture of progressive lymphedema, weight loss, lack of response to antibiotic therapy, and pleural effusion, metastatic malignancy was suspected. Accordingly, chest and abdominal computed tomography scan, skin biopsy, and blood investigations were undertaken.

Blood work up showed a complete blood count, organ function test and serum electrolytes within normal range, low total protein of 3.04 g/dl (reference range: 6.0–7.8), low albumin of 2.3 g/dl (reference range: 3.2–4.5), and high LDH of 307.0 U/l (reference range: 135.0–225.0). A computed tomography scan of chest and abdomen revealed bilateral pleural effusion, two pulmonary nodules in the left lower lobe, small ascites, and circumferential rectal wall thickening at the rectosigmoid junction.

Skin biopsy was taken from multiple sites and histologic sections from all sites show dermal and subcutaneous infiltrates of nests and sheets of signet ring cells with eccentrically located hyperchromatic nuclei and abundant eosinophilic to vacuolated cytoplasm in addition to frequent lymphovascular invasion (Fig. 2). Immunohistochemistry was carried out and the infiltrative epithelial cells tested positive for CK 7, CK 20, and CDX2 (Fig. 3) while negative for PSA and Hep Par-1. This is strongly suggestive of colorectal origin so colonoscopy with biopsy was done.

On colonoscopy visualized parts of rectum and sigmoid colon showed erythematous mucosa with mucosal erosions and friability from which biopsy was taken which ultimately revealed signet ring cell carcinoma (Fig. 3) confirming the diagnosis of cutaneous lymphangitis carcinomatosa secondary to colonic signet ring cell carcinoma.

The patient was ultimately sent to oncology and while he was being prepared for chemotherapy, he deteriorated fast and unfortunately died 3 months after the diagnosis. A summary of the timeline of the case is outlined in Figure 4.

Discussion

The incidence of cutaneous metastases has historically been reported between 0.7 and 9%; however, more recent studies have shown an increasing incidence that is closer to 5.3–10.4%[4]. The likelihood of developing cutaneous metastases depends on the primary cancer; breast cancer is the most common cause of cutaneous metastases overall, whereas lung cancer is the second-most common cause overall and the most common cause in men[5]. Only 4% of all cutaneous metastatic disease originates from colorectal cancers[7].

Although cutaneous metastasis typically signifies widespread disease and a poor prognosis, several cases of isolated rectal metastases to the skin without evidence of visceral disease have been reported[1]. Also, in a minority of patients, cutaneous disease may precede metastasis to other organs[1]. In these cases, it is imperative that the skin lesion be detected early, biopsied, and properly diagnosed[1]. Alternately, skin metastasis could be the first sign of an underlying advanced rectal cancer[1]. Skin metastases from colorectal cancer without visceral involvement are very rare and relate to short survival[7]. The average survival of a patient with cutaneous metastases varies from 3 to 18 months[7].

Cutaneous metastases from rectal carcinoma can reach the skin by four main routes[7]. Most commonly, tumor cells may spread to the skin by lymphatic and hematogenous disseminations[7]. However, direct extension from the underlying tumor to the skin or accidental implantation of tumor cells in surgical wounds and the surgical tract is also frequently reported[7].

The most typical sites of colorectal cancer metastasis are the lungs, bone, and liver[8]. Cutaneous metastases from rectal cancer are exceedingly rare and may appear as either an early or late

Figure 1. Picture showing grossly swollen scrotum and penis with cobbling of the overlying skin along with edema of the bilateral lower limbs.

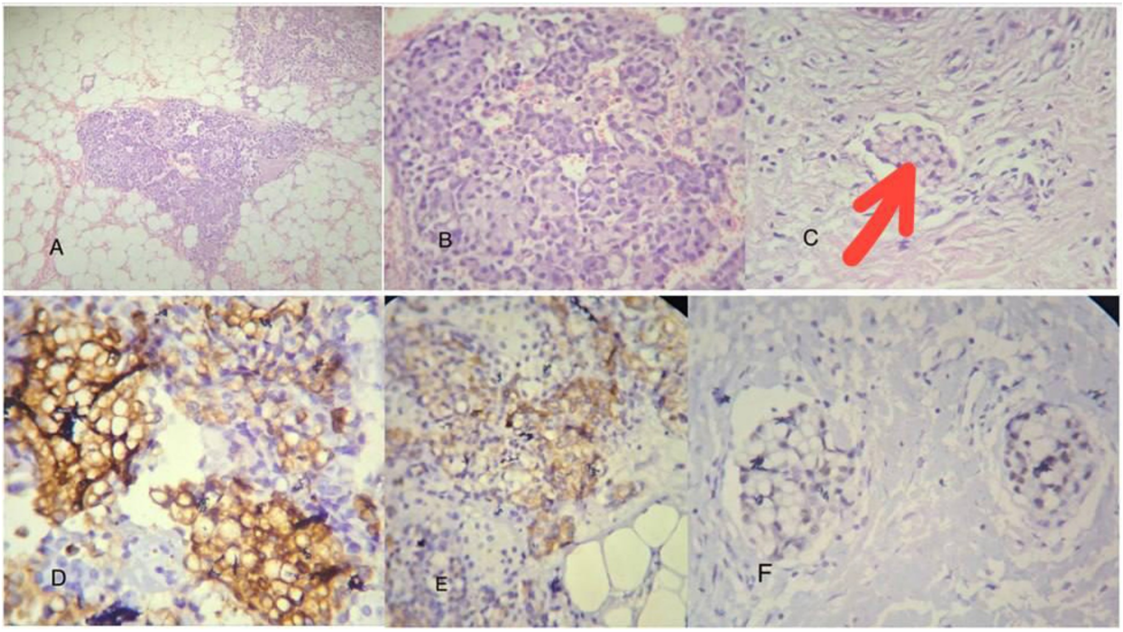

Figure 2. Picture showing histologic sections and immunohistochemistry staining from skin biopsy specimens. Picture A (Hematoxylin & Eosin, 10×) and picture B (Hematoxylin & Eosin, 40×) show subcutaneous adipose tissue infiltrated by malignant signet ring cells. Picture C (Hematoxylin & Eosin) shows lymphovascular invasion indicated by a red arrow. Pictures D, E, and F show that the tumor is positive for CK20, CK7, and CDX2, respectively. All controls (i.e., internal negative patients' tissue and external positive controls) show appropriate reactivity. Detection system used is Envision flex system, universal DAB detection with or without amplification kit on Dako auto stainer Link 48.

sign of the disease, often mimicking benign skin conditions[8]. The most frequent sites of cutaneous metastasis from colorectal cancer are abdomen followed by extremities, perineum, head, neck, and penis[9]. Diagnosing these metastases might be difficult, as they are rarely considered the primary clinical suspicion[8].

Signet-ring cutaneous metastasis is extremely rare, and usually occurs in the later stage of the disease, further signifying a poor prognosis[3]. The signet ring pattern defines a specific cell shape change, during which the cell nucleus is pushed to the periphery due to cytoplasmic accumulation of mucin, vacuoles, or inclusion bodies[10]. These lesions are often cutaneous or subcutaneous, normochromic or erythematous nodules, often asymptomatic, rarely presenting as inflammatory metastases[10]. In our case, the cutaneous secondaries manifested as an inflammatory metastasis, carcinomatous lymphangitis.

Cutaneous lymphangitis carcinomatosa, a rare form of cutaneous metastasis in less than 5% of cases, and occurs when malignant cells invade the dermis and the subcutaneous lymphatics[5]. The correlation between lymphoedema and cancer treatment is well known but secondary lymphedema caused by infiltration of the superficial dermal lymphatics is rare and resembles dermatosis[11]. Regardless of the primary tumor type, the prognosis for patients with cutaneous lymphangitis carcinomatosa and cutaneous metastases in general is poor, with demise typically occurring within months after diagnosis of skin involvement[5].

Skin metastasis is identified based on visual characteristics, histomorphology, and immunohistochemistry of the growth[8]. No single clinical presentation reliably diagnoses any form of cutaneous metastasis, as they can present in a variety of ways and may mimic other inflammatory or infectious processes[5]. It is suggested that the skin biopsy and immunohistochemical staining is not only helpful but essential for the diagnosis of SRCC and determining the primary cancer, which should be taken with no hesitation when a suspicion of cancer arises[3]. Like all cutaneous metastases, a punch or excisional biopsy is

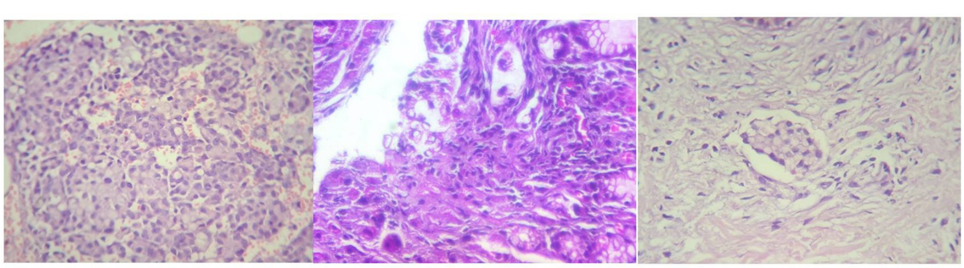

Figure 3. Histologic sections showing colonic mucosa infiltrated by malignant signet ring cells containing intracellular mucin, which displaces the nucleus to the side, with areas of lymphovascular invasion.

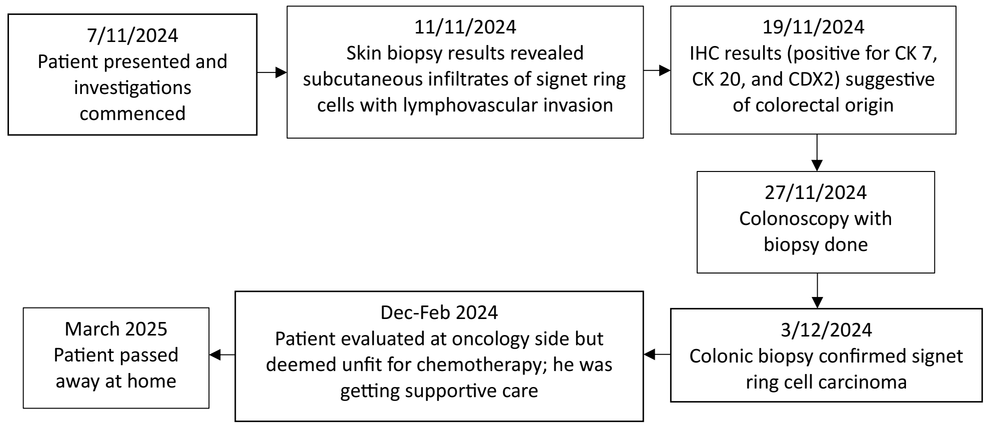

Figure 4. Timeline of the presented case. IHC, immunohistochemistry.

required to diagnose cutaneous lymphangitis carcinomatosa; the histology will resemble that of the primary tumor and immunohistochemistry may be required for confirmation[5]. A high index of suspicion is required to correctly diagnose cutaneous lymphangitis carcinomatosa, especially when the presentation is atypical[5].

Considering that the malignant cells will continue to synthesize tissue-specific proteins that can be identified by targeted immunostaining, these cells have conserved an adequate level of differentiation from the primary tumor[3]. In our case, the identification of signet ring cell morphology prompted immunohistochemical staining to evaluate for potential gastrointestinal malignancies. The tumor demonstrated immunopositivity for CK20, CDX2, and CK7. This immunoprofile, particularly the strong expression of CK20 and CDX2, supports a colorectal origin.

Lymphoedema caused by malignant infiltration is a chronic progressive disease with poor prognosis[11]. Even if the primary cancer has already been identified, correctly diagnosing cutaneous lymphangitis carcinomatosa can lead to adjustments of the patient's treatment plan that improve quality of life, as the lymphedema of cutaneous lymphangitis carcinomatosa is often symptomatic, hinders movement, and is psychologically distressing[5]. Adequate psychosocial support and terminal care is crucial, particularly at the last stage of the disease[11].

Conclusion

Although cutaneous metastases are relatively rare, their detection carries significant clinical importance. Their development is indicative of disseminated metastatic disease and is consequently associated with a markedly poor prognosis. Prompt recognition through careful physical examination is essential for effective management. Consequently, suspicious skin lesions should be considered potential indicators of an occult visceral malignancy, necessitating a high index of clinical suspicion even in patients without an established oncologic history.

Ethical approval

Ethical approval is not required for a case report publication from our institution.

Consent

Written informed consent was obtained from the patient's brother for publication of this case report and accompanying images. Consent was provided by next of kin due to the death of the patient. A copy of the written consent is available for review by the Editor-in-Chief of this journal on request.

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Author contributions

A.U.G.: drafting and writing of the manuscript, management of the patient; A.G.N.: revision and approval of the manuscript, management of the patient; A.S.T.: revision and approval of the manuscript, management of the patient; Y.G.Y.: reviewing of the pathology specimens and immunohistochemical analysis.

Conflicts of interest disclosure

The authors of this case report declare no conflict of interest.

Research registration unique identifying number (UIN)

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Data availability statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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